Lecture 28.05.2026

Van der Pol equation

$$\ddot{q} + q - \varepsilon(1 - q^2)\dot{q} = 0, \quad \varepsilon > 0$$

$\exists$ asymptotically stable limit cycle

[...]

$$g_\varepsilon(E_0) := E_0 + 2\pi\varepsilon E_0\left(1 - \frac{E_0}{2}\right) + \varepsilon^2 h(E_0, \varepsilon)$$

g_ε admits a unique fixed point

with

$$h(E, \varepsilon) := \int_{2\pi}^{T_\varepsilon} \frac{ds}{\varepsilon} F(E, -s) + \frac{1}{\varepsilon}\int_0^{T_\varepsilon} \left(F(E(s), \theta(s)) - F(E, -s)\right) ds$$

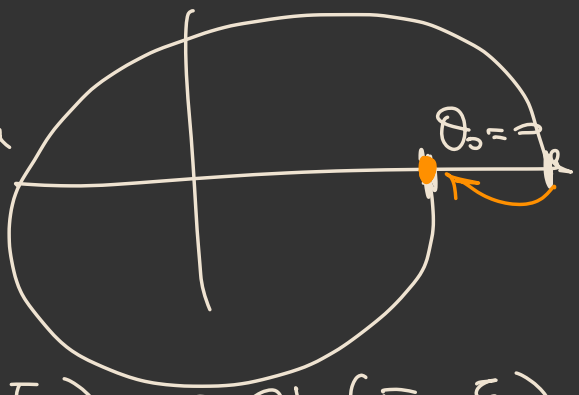
$T_\varepsilon \in (0, 3\pi)$ and $F(E, \theta) = 2E\sin^2\theta(1 - 2E\cos^2\theta)$

Proving $\exists$ fixed point for $g_\varepsilon(E_0)$ is equiv. to

$$k(E_0, \varepsilon) := 2\pi E_0\left(1 - \frac{E_0}{2}\right) + \varepsilon h(E_0, \varepsilon) \overset{!}{=} 0$$

⚠ $\varepsilon > 0 \Rightarrow \exists!$ sol for $\theta_0 = 0$ that is at $E_0 = 2$

Infact, by implicit fnct thm



$$\frac{\partial k}{\partial E_0}(E_0, \varepsilon) = 2\pi(1 - E_0) + \varepsilon \underbrace{\frac{\partial h}{\partial E_0}(E_0, \varepsilon)}_{\text{from the estimates for } \frac{\partial h}{\partial E_0}}$$

$$\Rightarrow \frac{\partial k}{\partial E_0}(2, 0) = -2\pi \neq 0$$

$\Rightarrow$ by implicit fnct thm

$\exists\, (-\varepsilon_0, \varepsilon_0)$, $\exists\, \varepsilon \longmapsto \bar{E}(\varepsilon)$ s.t.

$\varepsilon \in (-\varepsilon_0, \varepsilon_0)$

$$k(\bar{E}(\varepsilon), \varepsilon) = 0 \quad \text{with } \bar{E}(0) = 2$$

$\Rightarrow \exists!$ isolated fixed point

$g_\varepsilon(\bar{E}(\varepsilon)) = \bar{E}(\varepsilon)$, corresponding to a limit cycle close to the circonference at radius $\sqrt{2E_0} = 2$.

In order to prove that the limit cycle is asymptotically stable, we compute

$$g_\varepsilon'(E_0) = 1 + 2\pi\varepsilon\left(1 - E_0\right) + \varepsilon^2 \frac{\partial h}{\partial E_0}(E_0, 0)$$

$$g_\varepsilon'(2) = 1 - 2\pi\varepsilon + O(\varepsilon^2) < 1$$

$\Rightarrow$ the limit cycle is asymptotically stable

□

PDE theory - models.

$y = y(t)$, $t > 0$

$u = u(t,x)$, $t > 0$, $x \in \Omega \subseteq \mathbb{R}^N$

↝ reaction - diffusion equations

* time evolution part
* diffusion effect
* reaction in the environment

(*) $\partial_t u - d\Delta u = f(u)$, $(t,x) \in (0,+\infty) \times \Omega$

evolution part

$N = 1$ $\Delta u = \partial^2_{xx} u$

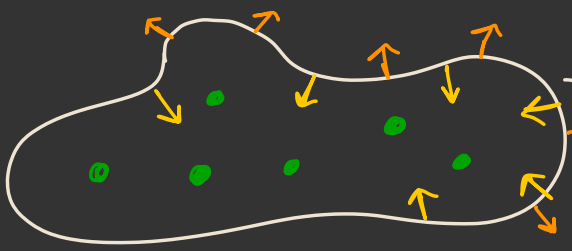
$d > 0$ linear diffusion coefficients

$f(u)$: reaction part

$f(u) = u(1-u)$ ↝ (*) Fisher-KPP model

Ω bounded domain $\Omega \subset \mathbb{R}^N$

$\rightsquigarrow$ boundary condition



$\Omega \subset \mathbb{R}^N$, $N \geq 1$

$\partial\Omega \in C^2$

1) Homo Dirichlet boundary condition
generous

$$u(t,x) = 0 \quad , \quad (t,x) \in (0,+\infty) \times \partial\Omega$$

(Non-homog. $u(t,x) = g(x,u)$, $x \in \partial\Omega$)

Unfavorable conditions for living in Ω

2) Homogeneous Neumann boundary condi

$$\nabla_x u(t,x) \cdot \nu = 0, \quad (t,x) \in (0,+\infty) \times \partial\Omega$$

$\nu = \nu(x)$ outward unit normal

(Non-homo Neumann bound cond)

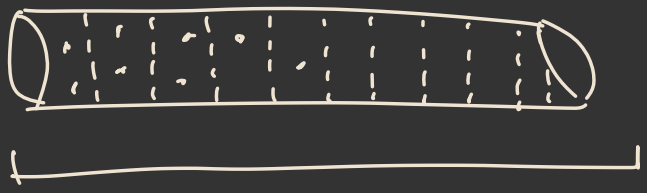
The ecosystem is isolated.

(Formal) Derivation of diffusion eq.

$$\partial_t u - d\Delta u = 0$$

Notes on Noodle

$\varepsilon = L/N$ N



$L > 0$

$\varepsilon \to 0$

We deal with PDEs systems of 2 equations.

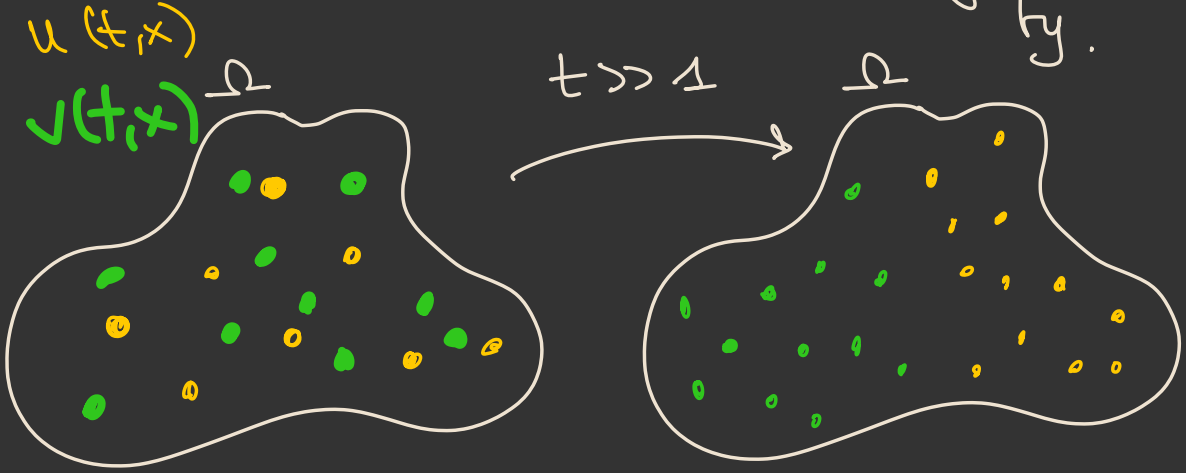
$u = u(t,x) \geq 0$

$v = v(t,x) \geq 0$

$$(RD) \begin{cases} \partial_t u - d_u \Delta u = f(u,v), & (t,x) \in (0,+\infty) \times \Omega \subset \mathbb{R}^N \\ \partial_t v - d_v \Delta v = g(u,v), & (t,x) \in (0,+\infty) \times \Omega \end{cases}$$

$d_u, d_v > 0$ diffusion coefficients.

GOAL : Stability analysis of (RD) with interest in Turing instability.



$\Rightarrow$ Segregation of species occurs (Turing instability)

in Morphogenesis (1954)

Why instability: (homogeneous)

1) Identification $\sqrt{}$ steady state $(\bar{u}, \bar{v})$

2) $d_u = d_v = 0 \rightsquigarrow$ associated ODE system to (RD)

$(\bar{u}, \bar{v})$ linearly stable

3) $d_u, d_v \neq 0$

$(\bar{u}, \bar{v})$ linearly unstable

Turing instability

$$(RD)\begin{cases} \partial_t u - d_u \Delta u = f(u,v), & (t,x) \in (0,+\infty)\times\Omega \\ \partial_t v - d_v \Delta v = g(u,v), & (t,x) \in (0,+\infty)\times\Omega \end{cases} \quad \Omega \subset \mathbb{R}^N$$

with IC

$$u(0,x) = u_0(x) = u_0 > 0, \quad x \in \Omega$$

$$v(0,x) = v_0(x) = v_0 > 0, \quad x \in \Omega$$

and BD

$$\nabla_x u \cdot \nu = \nabla_x v \cdot \nu = 0, \quad (t,x) \in (0,+\infty)\times\partial\Omega$$

STEP 1)

$\exists\, (\bar{u}, \bar{v}) \in \mathbb{R}^2_+$ s.t. $\begin{cases} f(\bar{u},\bar{v}) = 0 \\ g(\bar{u},\bar{v}) = 0 \end{cases}$

↪ homo steady states for (RD).

We consider small spatial perturbation

$$u = \bar{u} + \varepsilon \tilde{u}, \quad |\varepsilon| \ll 1.$$
$$v = \bar{v} + \varepsilon \tilde{v}$$

STEP 2) $(d_u = d_v = 0)$

$$\partial_t \begin{pmatrix} \tilde{u} \\ \tilde{v} \end{pmatrix} = J(f,g)\Big|_{(\bar{u},\bar{v})} \begin{pmatrix} \tilde{u} \\ \tilde{v} \end{pmatrix}$$

$$J(f,g)\Big|_{(\bar{u},\bar{v})} = \begin{pmatrix} \partial_u \bar{f} & \partial_v \bar{f} \\ \partial_u \bar{g} & \partial_v \bar{g} \end{pmatrix}$$

$\bar{f} = f(\bar{u},\bar{v})$

$\bar{g} = g(\bar{u},\bar{v})$

$$\operatorname{tr} J(\bar{f},\bar{g}) = \partial_u \bar{f} + \partial_v \bar{g} \overset{!}{<} 0$$
$$\det J(\bar{f},\bar{g}) = \partial_u \bar{f}\, \partial_v \bar{g} - \partial_u \bar{g}\, \partial_v \bar{f} \overset{!}{>} 0$$

$\Rightarrow (\bar{u},\bar{v})$ is linearly stable for associated ODE system of (RD).

STEP 3) $d_u, d_v \neq 0$

$$u(t,x) = B(t)\,W(x)$$

$B(t)$ → linear ODE

$W(x)$ → eigenvectors of eigenvalues △

